being consistent with all the luminosity being emitted in the photosphere ($\Gamma_w \sim 2.5$, a passive disc: Petrucci et al. 2018), to being much steeper as expected for increasing dissipation in the warm disc itself.

Overall, our model fits show a clear and systematic evolution in the SED, which corresponds to a change in the accretion structure itself, transitioning from a disc(-like) dominated state at high $\dot{m}$ to one dominated by the X-ray plasma at low $\dot{m}$ (much like the low-hard state in BHBs.

4 DISCUSSION AND CONCLUSIONS

All the SEDs considered here are consistent with the standard outer disc component appearing below a roughly constant temperature of $\sim 4500-6500$ K, which is also the temperature region where Hydrogen goes from being mostly ionised (above T_{hi}) to mostly neutral, perhaps providing insight into the origin of the warm Comptonising disc-like structure.

Disc temperatures around $1-5\times10^4$ K predict local SEDs peaking in the UV/EUV. Stars at these temperatures (O stars, Wolf-Rayet, and luminous blue variables) have significant winds powered by UV line-driving (e.g Puls et al. 2008; El Mellah et al. 2017). This can translate to AGN discs, powering radiatively driven outflows, through e.g UV line driving (Proga et al. 2000; Proga & Kallman 2004). Strong winds from an accretion disc naturally lead to significant mass-loss, which can produce a distinct change in the disc properties themselves (Laor & Davis 2014). Our observed T_{hi} is an order of magnitude below these characteristic UV wind temperatures, making the warm disc region unlikely to be connected to this process.

Instead, there is a dramatic change in the disc opacity at the Hydrogen ionisation threshold at around 6000 K (Cannizzo & Reiff 1992), leading to a local instability (Cannizzo 1992). In stellar mass BHB discs, which cross this temperature at large radii where there is not much power from accretion, the disc opacity changes so dramatically that this local instability triggers a global limit cycle, within the disc, so it transitions between outbursts and longer periods of quiescence (Lightman & Eardley 1974; King & Ritter 1998; Lasota 2001; Dubus et al. 2001). The lower temperature discs in AGN mean that SMBHs cross this temperature range at much smaller radii (scaled in terms of the gravitational radius) where there is significantly more power emitted. Thus this instability might be expected to have an even more dramatic impact on the disc structure. However this is not generally seen in the simulations of this so far (Burderi et al. 1998; Siemiginowska & Elvis 1997; Grzędzielski et al. 2017), but AGN discs are also in the regime where radiation pressure dominates, and we speculate that the non-linear outcome of these two instabilities acting together holds the key to understanding the origin of the warm disc region in AGN.

In addition it is quite clear from Fig. 5 that while the optical luminosity changes by roughly 3 orders of magnitude, the hot X-ray component only appears to change by a factor of a few. While our model explains this as the flow becoming increasingly dominated by the X-ray corona as $\dot{m}$ decreases, we do not have a good physical reason for *why* the flow seemingly smoothly transitions into an X-ray corona. Though, we note that the behaviour of the X-ray power is roughly consistent with the toy model of Lusso & Risaliti (2017), which is based off Svensson & Zdziarski (1994) (and expanded upon in Merloni & Fabian 2002 and Arcodia et al. 2019) and states that the X-ray corona only becomes relevant once gas pressure begins to dominate over radiation pressure, setting a mass-accretion rate dependent size-scale of the X-ray corona. Alternatively, it appears that the X-ray always emits at $\sim 1-2$ % of the Eddington luminosity, which is also roughly the maximal luminosity for an ADAF flow (Narayan & Yi 1995; Yuan & Narayan 2014). We speculate that perhaps this is showing that AGN always have a maximal ADAF present; contrary to BHBs.

Regardless of the origin of the warm disc-like region and its transition into the X-ray corona, it is fundamentally clear that it depends strongly on $\dot{m}$. The clear smooth transition in the SEDs from a disc dominated state to one dominated by the X-ray, is most easily explained by a change in the accretion structure itself, as shown previously for some rare changing-look (changing-state) AGN (Noda & Done 2018). This change cannot be due to obscuration, as the X-rays are clearly unobscured (by selection - see also the discussion in Appendix B). The inner accretion flow in AGN is relatively compact, and so we would expect any obscuring medium present in the UV to intercept the X-rays too. Of course, there could be some low column-density dusty/gas on large scales that would contribute to the reddening of the optical/UV continuum, but would not be seen in our X-ray data (e.g Czerny et al. 1995). However, we find it unlikely that this could cause sufficient reddening to give the extremely red continua seen in the lowest three luminosity bins, while remaining X-ray unobscured. In addition the transition from a strong soft X-ray excess in the highest three luminosity bins, to a single non-thermal X-ray component in the lowest three bins, strongly suggests a collapse of the structure responsible for the soft excess as $\dot{m}$ reduces rather than increased obscuration.

Along with the collapse of the disc, and thus the EUV continuum, we expect to see a reduction/loss of the broad emission lines, due to a reduction/loss of the ionising photons. This is entirely analogous with the changing-look phenomenon (Noda & Done 2018; Ruan et al. 2019), except now it is a generic feature of the AGN population rather than limited to a few rare objects. In some sense this has been previously hinted at, with studies showing a sharp drop in the number of broad line AGN below 0.02 $L_{\rm Edd}$ (Trump et al. 2011; Mitchell et al. 2023), which is both where the changing-look transition is known to occur and where our sample shows a loss of the EUV continuum. This can easily be tested by matching our HSC-eROSITA sample to one of the spectroscopic surveys (e.g SDSS York et al. 2000 or DESI DESI Collaboration et al. 2016). This is beyond the scope of our current paper, where we limit ourselves to the SED, but will be addressed in future work.

The more obvious limitation within our work is the data-gap between the optical/UV and the X-ray. Our modelling *predicts* substantial power emitted in the EUV at high $\dot{m}$ and the loss of this power at low $\dot{m}$. However, this is predictive, and entirely based on what the model requires to bridge the gap between the optical/UV and X-ray, while maintaining energy conservation. It is entirely plausible that alternative accretion flow models could predict entirely different behaviour in the EUV gap, which would fit our data just as well. In this sense additional data in the near-UV (NUV) and far-UV (FUV) could work to strengthen (or falsify) our conclusions. This will be the topic of a future paper, using archival GALEX data (Kang et al. (in prep)).

Nonetheless, there is a clear drop in the blue optical continuum from the warm disc at $\dot{m} \lesssim 0.02$. This is clearly a generic feature in the AGN population, and not limited to local low-luminosity AGN (LINERs) (Ho 1999) and a few rare individual changing-state AGN (Noda & Done 2018). There is a real, and quite abrupt change in the accretion structure, driven by the changing Eddington ratio, which leads to the complete loss of the ionising EUV emission, and hence the characteristic broad emission lines. This has a significant impact on how we identify AGN through cosmic time. In an era where the James Webb Space Telescope is breaking new ground with the

discoveries of quasars at cosmic dawn, any inference for the overall population requires we understand the nature of the accretion flow itself. This study reveals a systematic change in the accretion structure; the collapse of the accretion disc; between high and low luminosity AGN, impacting what we infer about the general AGN population, structure and growth through cosmic time.

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This work made use of the following PYTHON modules: ASTROPY (Astropy Collaboration et al. 2013, 2018, 2022) and NUMPY (Harris et al. 2020). Additionally, the plots were made using a combination of MATPLOTLIB (Hunter 2007) and SEABORN (Waskom 2021).

DATA AVAILABILITY

The optical HSC-fluxes used in this paper are from Li et al. (2024), and the corresponding catalogue can be found at: https://member.ipmu.jp/john.silverman/HSC/.

The X-ray spectra are part of the eROSITA eFEDS early data-release AGN catalogue, as described in Liu et al. (2022). This is available through: https://erosita.mpe.mpg.de/edr/eROSITAObservations/Catalogues/.

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APPENDIX A: IMPORTANCE OF THE HSC HOST-AGN DECOMPOSITION

The results of our study rely on confidently extracting the AGN emission from the host galaxy emission, especially for the low luminosity bins. This is clear in Fig. A1, where we show the optical part of the stacked SEDs for both the de-composed AGN fluxes and the non-decomposed total fluxes for four of the luminosity bins ($\nu L_{3500} \in [42.5, 42.9]$, $[43.3, 43.7]$, $[44.1, 44.5]$, and $[44.9, 45.3]$ ergs/s). In all panels, except the highest luminosity bin, we clearly see that the non-decomposed SED gives the wrong optical luminosity; from an error of a factor few to an order of magnitude. Hence an attempt at this study with no host galaxy decomposition would give the wrong estimate of the system energetics, over predicting the contribution from a disc (especially at low luminosity), and hence arriving at the wrong conclusion on the balance between the X-ray and optical emitting parts of the accretion flow.

As well as systematically overestimating the normalisation, we see in Fig. A1 that the non-decomposed SEDs can give the wrong spectral shape (or slope). This is especially clear in the bin $\nu L_{3500} \in [43.3, 43.7]$ ergs/s, where the non-decomposed SED is clearly decreasing in power with frequency (energy), while the isolated AGN SED is more or less flat.

The AGN and host galaxy fluxes are decomposed using a 2D optical image decomposition method, as described in Li et al. (2024). For details on how the method works we refer the reader to Li et al. (2021a); Ding et al. (2021).

APPENDIX B: CRITICALLY TESTING FOR OBSCURATION IN THE X-RAY STACKS

We have based our study on the assumption that all the sources within our stacks are in fact unobscured, as highlighted in our selection criterion of $N_{H,\mathrm{X-ray}} < 10^{22}\,\mathrm{cm}^{-2}$. However, at this threshold N_H and spectral slope will be degenerate in some of the eROSITA sources Liu et al. (2022), leading to uninformed N_H measurements. Hence, there is a chance that some of the sources contributing to our stacks are in fact obscured. In the stacks presented in this paper (see Fig. 5), this possibility mostly concerns the lowest three luminosity bins, which appear to be characterised by a single power-law. The higher bins cannot be heavily obscured on average, as they display a soft X-ray excess, which would be wiped out if the average column-density in these bins were above $10^{22}\,\mathrm{cm}^{-2}$. In this appendix we therefore test the three lowest luminosity bins, to ensure that these results are convincingly due to a change in accretion state rather than spurious obscuration contaminating our sample.

We start by taking the eROSITA stacks from the three lowest luminosity bins from Fig. 5 (i.e $\nu L_{3500} \in [42.1, 42.5]$, $[42.5, 42.9]$, and $[42.9, 43.3]$). These initially appear as being characterised by a clear single power-law component. We apply a simple absorbed power-law model to these data, using the XSPEC model PHABS*POW, to test maximally allowed average column-density in the sample. These results are shown in Fig. B1, where we show $\Delta\chi^2$ curves calculated by stepping the fit through fixed N_H values (solid coloured lines). It is immediately clear that on average